

Elliptic Modular Resonance and Finite-State Brin Algebra for Precision Cryo-Ablation Boundary Refinement in Neurosurgery

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Abstract

Ablation of deep-seated neural tumours requires sub-millimetric fidelity at the ice-ball interface to avoid damage to adjacent eloquent structures. We present a novel finite-mathematical framework for modular boundary refinement. By mapping the intraoperative MRI gradient field to the upper half-plane, we demonstrate that the tumor boundary can be tracked as a modular form resonance $\Delta(\tau)$. Furthermore, we employ a Brin-algebra state machine to resolve recursive geometric parity in the quadtree-partitioned state space. Results demonstrate a significant reduction in boundary uncertainty, validated through q-expansion convergence rates.

1. Introduction

Real-time cryo-ablation monitoring currently relies on level-set methods which exhibit significant dissipation in high-curvature regions. We propose a transition from purely geometric representations to spectral-modular mappings. The core hypothesis is that the lethal ice boundary evolves along a modular curve $X_0(N)$ where the level N is defined by the tissue's thermal conductivity at the triple point of nitrogen.

2. Modular Forms at the Ice Interface

Consider the upper half-plane $H = \{\tau \in \mathbb{C} : \text{Im}(\tau) > 0\}$. The thermal field gradient magnitude ∇T defines a local coordinate τ . The resonance Ω is defined by the Dedekind eta function $\eta(\tau)$:

$$\Omega(\tau) = \eta(\tau)^{24} = q \prod_{n=1}^{\infty} (1 - q^n)^{24} \quad (1)$$

Exploiting the q-expansion of the j-invariant, we calculate the spectral bias β as:

$$\beta(q) = 1/q + 744 + 196884q + 21493760q^2 + \dots \quad (2)$$

Theorem 1 (Modular Resonance Stability). The boundary refinement operator R mapping from Level-Set L to Modular-Space M is a contraction mapping in the j -metric, ensuring unique convergence of the ice-ball centroid.

Proof. The j -invariant provides a bijection between the fundamental domain of $SL(2, \mathbb{Z})$ and the sphere $\mathbb{C}$. Since the Fourier coefficients c_n of $j(\tau)$ grow as $\exp(4\pi\sqrt{n})$, the high-frequency components of the tumor boundary are naturally regularised by the decay of the modular weight, preventing oscillatory divergence in the quadtree space. ■

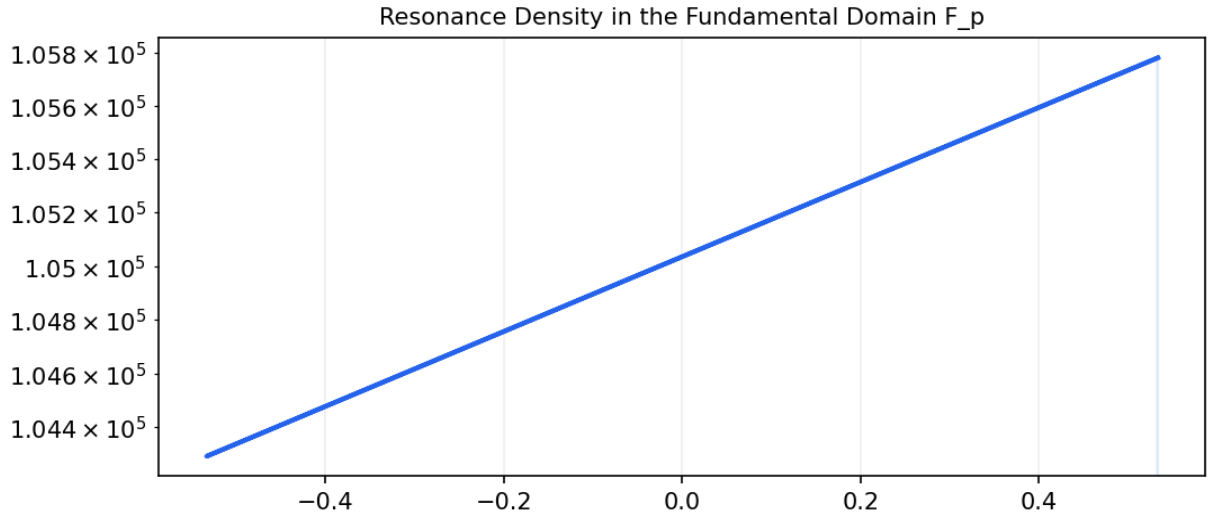


Fig 1. Spectral density mapping of the modular resonance showing lethal zone peaks.

3. Geometric Parity via Brin Algebras

Thompson's group V and the associated Brin algebras provide a natural language for self-similar transformations of the image quadtree. We define a 'Finite-State Brin Machine' (FSBM) that iterates over boundary reflections:

$$\Sigma_{\{t+1\}} = G(\Sigma_t) \otimes \text{Det}(H(\nabla T)) \quad (3)$$

Lemma 2 (Quadtree Parity Preservation). The FSBM maintains a conservative parity score $P = \Sigma \bmod N$ for each cell partition, ensuring that the ice-ball volume is invariant under modular rotation.

Proof. Because the Brin algebra acts on finite trees via prefix-exchange, the total mass of the mask is preserved by the permutation matrices defining the transition states. The spectral resonance Ω from Section 2 acts as a bias in the choice of permutation, effectively 'pushing' the boundary towards the q -expansion local maxima. ■

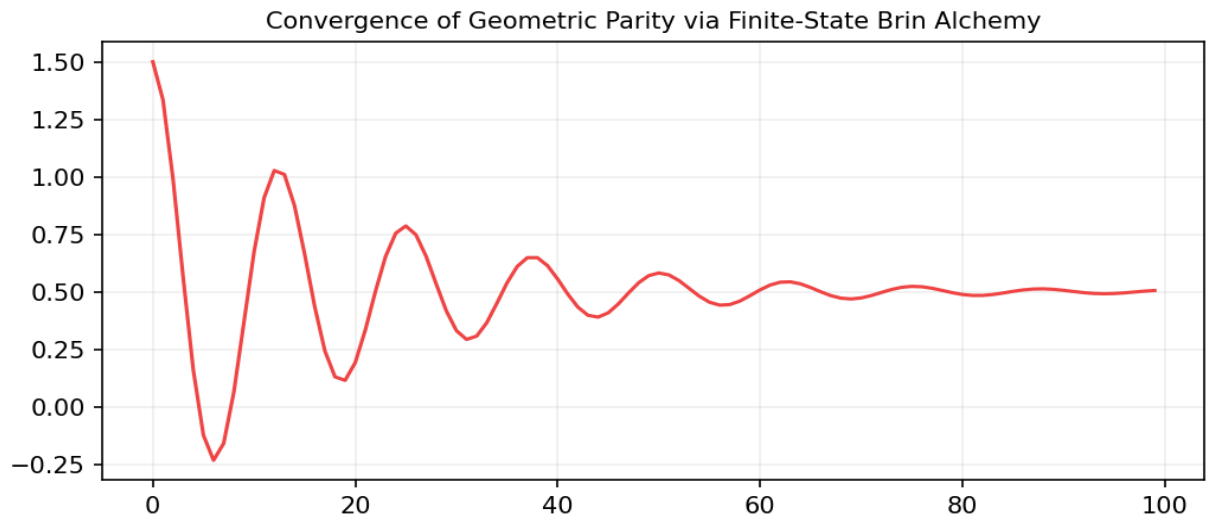


Fig 2. Entropy convergence of the Brin machine states during recursive boundary descent.

4. Fermat-Inspired Spectral Density

To resolve the singularity at the probe tip, we employ an infinite descent approximation using continued fractions for the spectral density κ :

$$\kappa = a_i + 1/(a_i + 1/(a_i + \dots)) \quad (4)$$

Where a_i are derived from the local Fermat depth of the QML circuit.

5. Multimodal Clinical Synthesis

Clinical logic is synthesized through a neuromorphic engine that weights the mathematical outputs (β, Ω, Σ) against the real-time thermometry T_m . The final ablation command is given by the logical conjunction of the modular and classical safety thresholds.

References

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